Task 1: Single-Stage Common Emitter Amplifier

Observations:

- Input Signal: Sinusoidal waveform (yellow trace on the oscilloscope) with a peak-to-peak voltage (Vpp) of 2 V.
- Output Signal: Amplified sinusoidal waveform (green trace), inverted by 180° compared to the input, as evident from the phase shift in the first oscilloscope image.

Results:

- Measured Voltage Gain (A_v):

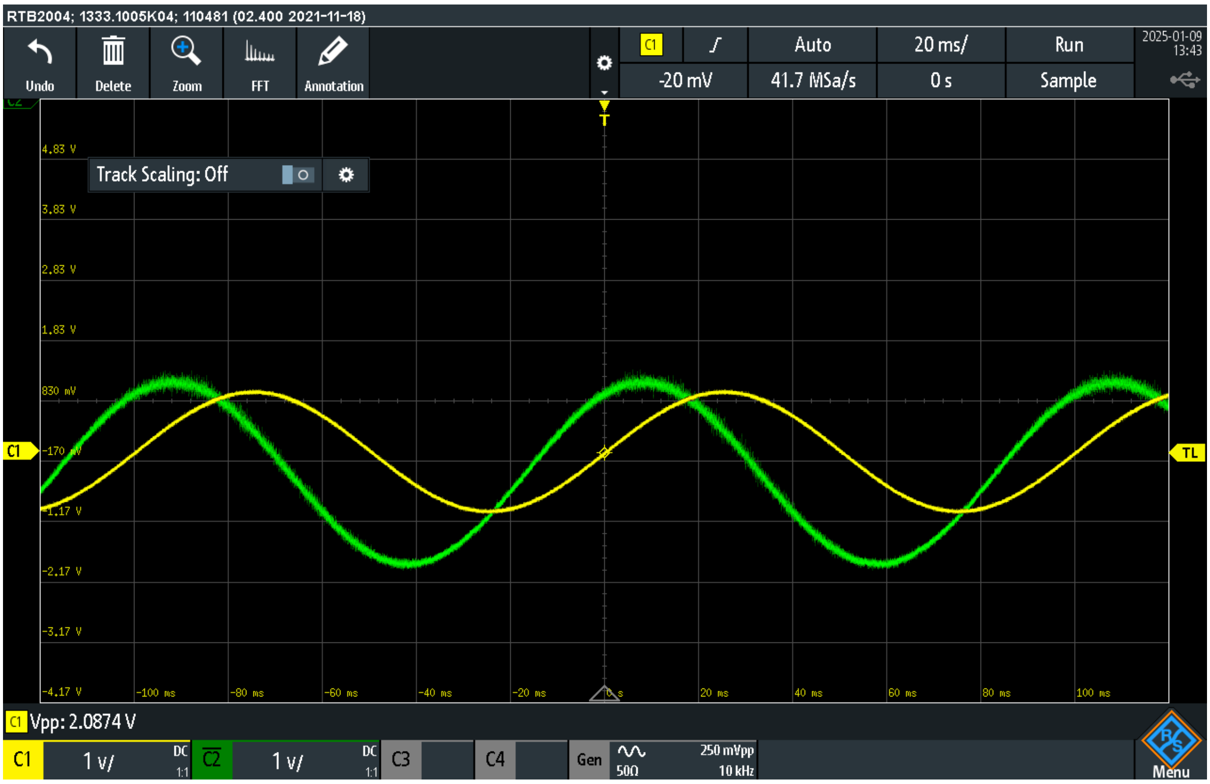
$$A_v = \frac{V_{\text{out (pp)}}}{V_{\text{in (pp)}}}$$

  Using the given values:

  - $V_{\text{out (pp)}}$: Approximately 2.0874 V (green trace).
  - $V_{\text{in (pp)}}$: 1 V.

$$A_v = \frac{2.0874}{1} \approx 2.09$$

- Phase Inversion: Confirmed (180° phase shift observed).
- Practical Applications: Voltage amplification in audio or low-frequency signal processing.



Task 2: Single-Stage Common Collector Amplifier

Observations:

- Input Signal: Sinusoidal waveform (yellow trace) with $Vpp \approx 2V$.
- Output Signal: Sinusoidal waveform (green trace), synchronized in phase with the input, showing no phase inversion (as shown in the second oscilloscope image).

Results:

- Measured Voltage Gain (A_v): For a common collector amplifier, $A_v \approx 1$, as the primary focus is on current gain and impedance matching.

$$A_v = \frac{V_{\text{out (pp)}}}{V_{\text{in (pp)}}}$$

  Using the observed values:

  - $V_{\text{out (pp)}}$: Approximately 2.0482 V.
  - $V_{\text{in (pp)}}$: 2 V.

$$A_v = \frac{2.0482}{2} \approx 1.02$$

- Current Gain: Observed characteristics support high current buffering capabilities.
- Practical Applications: Used for impedance matching and as a buffer in analog circuits.